

Homework Section 15.3

1. Evaluate the iterated integral.

a) $\int_1^2 \int_x^2 xy dy dx$

b) $\int_0^{\pi/4} \int_0^{\cos \theta} \sin \theta dr d\theta$

2. Evaluate the double integral.

a) $\iint_R \frac{y}{x^3+2} dA, R = \{(x, y) | 1 \leq x \leq 2, 0 \leq y \leq 2x\}$

b) $\iint_R e^{x/y} dA, R = \{(x, y) | 0 \leq x \leq y^3, 1 \leq y \leq 2\}$

c) $\iint_R y\sqrt{x^2 - y^2} dA, R = \{(x, y) | 0 \leq y \leq x, 0 \leq x \leq 1\}$

d) $\iint_R x \cos y dA, R$ is bounded by $y = 0, y = x^2, x = 1$

3. Find the volume of the given solid:

a) Under the plane $x + 2y - z = 0$ and above the region bounded by $y = x$ and $y = x^4$.

b) Under the surface $z = xy$ and above the triangle with vertices $(1, 1)$, $(4, 1)$, and $(4, 7)$.

c) Bounded by the cylinder $y^2 + z^2 = 4$ and the planes $x = 2y, x = 0, z = 0$ in the first octant.

4. Evaluate the integral by reversing the order of integration:

a) $\int_0^1 \int_{3y}^3 e^{x^2} dx dy$

b) $\int_0^1 \int_{x^2}^1 x^3 \sin(y^3) dy dx$